Collision and Scattering for Process Molecules

Graded where-clauses beyond the well-mixed solution

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Abstract

The structural congruence of the rho calculus makes a large parallel composition a multiset, and a multiset is a perfectly stirred beaker: any receiver may meet any matching message, and nothing but multiplicity distinguishes one configuration from another. This is exactly the well-mixed hypothesis behind Gillespie's stochastic simulation, which the graded where-clause already realises. This note asks what must be added if the metaphor of a process term as a molecule is to be taken further — far enough to speak of bonds, of viscosity and, eventually, of turbulence.

The note records a design path and the construction it arrived at. A *location* is a splitting $\langle K, t\rangle$ of a term, pointed to by $K[t]$. A race is two terms bidding for the same location, which is a critical pair in the sense of Knuth and Bendix. Ordinary completion would join or orient the peak; we instead *name* it. A collision is the case in which both bidders arrive, and its outcome is a substitution produced by a scattering kernel into the several holes of the receiver's continuation. Once receipts may destructure their payload by pattern, the kernel need not be primitive: matching the receiver's pattern against the parallel composite of the arrivals is a canonical kernel, k-body collisions are matching against k-fold composites, and the total firing rate at a site acquires a series $\sum_k \binom{n}{k} w_k$ whose k-th coefficient plays the role of the k-th virial coefficient. The rho calculus is the truncation at $k = 1$, the ideal gas.

We leave receipts unmarked and count the cost. A receipt's shape decides whether it can collide at all; the cross-section decides whether it does; and the translation back to rho is the identity wherever the cross-section is zero. The translation itself — a lock and one self-join per split of the pattern — is correct only after a correction we did not anticipate: each translated candidate must be weighted by the inverse of its *fibre multiplicity*, a contingency-table count that the symmetry factor $1/k!$ alone does not capture, but which is computable locally and is therefore admissible in a where-clause. Over 3,000 random instances the corrected translation agrees with the kernel outcome by outcome to 1.6×10^{-15}; the uncorrected one overcounts by a factor of up to 24.

Three worked examples close the note: the small kernels (association, which is bonding; exchange; fission); collisional activation in the Lindemann–Hinshelwood mechanism, whose fall-off curve the kernel reproduces; and, as the example that is more than a toy, a lattice gas whose shear viscosity is produced by the collision kernel. The two-body kernel on a square lattice has a conserved quantity on every row and its shear wave never decays; the two-body kernel on a hexagonal lattice still has one on every horizontal line, and its shear wave stalls at the predicted two thirds; adding the three-body term removes the invariant, the shear wave decays, and the measured viscosity falls with the collision parameter, as a mean-free-path term in $1/\lambda$ on top of a contribution from the lattice itself.

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1 Introduction

1.1 The question

In the rho calculus the parallel composition of processes is associative and commutative, with 0 as unit. A population of process terms in parallel is therefore a multiset, and it has no positions: there is no sense in which one message is nearer to one receiver than to another. The equations were originally intended to model a well-mixed solution, and they do so exactly. Gillespie's derivation of the chemical master equation rests on the same hypothesis — reaction propensities depend on molecule counts alone — and the graded where-clause of the

companion notes [6, 17, 20] gives a rho program precisely Gillespie's semantics when the clause is valued in $\mathbb{R}_{\geq 0}$: candidates are counted as selections from the multiset, weighted by the clause, and resolved as the jump chain of a continuous-time Markov chain [5].

But a well-mixed solution has no viscosity, no shear and no eddies. If the metaphor of a process term as a molecule is to be taken more seriously — far enough to speak of bonds, of transport and eventually of turbulence — the calculus must say more than the multiset says. The question this note pursues is what, minimally, must be added, and whether the graded where-clause can carry it.

1.2 What a fluid needs that a beaker does not

Three things are erased by the multiset and needed by any fluid. *Position*: locality, so that only neighbours can react. *Momentum*: viscosity is the diffusion of momentum, and turbulence is a regime of a velocity field; without a momentum carried by the molecule there is nothing for either to be about. *Time*: the graded semantics over $\mathbb{R}_{\geq 0}$ is confined to the jump chain ([17], remark "Gillespie, precisely"), while transport coefficients live in rates. The construction of this note supplies the second directly, gives the first a syntactic carrier (although not, yet, an emergent one), and works with rates throughout.

1.3 What the note does

Section 2 records the design path: making the equations into weighted rewrites, letting geometry emerge from interaction, and the problem of bonding. Section 3 recasts a race as a critical pair. Section 4 introduces the collision kernel: two terms that race for the same location both arrive, and their scattering is a substitution into the receiver's holes. Section 5 makes the kernel canonical using destructuring patterns and derives the virial series. Section 6 leaves receipts unmarked and counts the cost. Section 7 maps the kernel back to rho and isolates the fibre-multiplicity correction. Section 8 gives the worked examples, Section 9 separates what is shown from what is open, and Appendix A documents the computations. Every number in the note is produced by `collide.py`, which ships alongside it and uses only the Python standard library.

1.4 Conventions

Quotation is $@P$ and dropping is $*x$. A send is $x!(P)$; a linear receipt is $\texttt{for}(@\pi \leftarrow x)R$ and a persistent one $\texttt{for}(@\pi \Leftarrow x)R$, where π is a pattern (Section 5). Structural congruence $\equiv$ contains the monoid laws for $|$ and 0 and the equation $*@P \equiv P$, and names are equal when their quoted processes are congruent. In a pattern a variable v matches a process S and binds the *name* $@S$, so the continuation recovers the process as $*v$; this is the convention of the 2005 calculus [15], in which receipts bind names. Graded where-clauses take values in a commutative semiring $\mathbb{V}$; except where stated, $\mathbb{V} = \mathbb{R}_{\geq 0}$. A *site* is a channel together with the receipts and messages on it.

2 The design path

The construction of Sections 3–7 was not the first idea, and the ideas it displaced explain its shape. This section records them briefly.

2.1 Two routes

There are two ways to make a population of process terms behave like a fluid. On the *phenomenological* route viscosity is a parameter: a graded damping on transport rates over a well-mixed or compartmentalised base, in the manner of the reaction–diffusion master equation [4]. This is cheap and sits within the existing machinery. On the *mechanistic* route viscosity is an output. Lattice-gas automata are the precedent [11, 2, 3]: a discrete rewrite system recovers the Navier–Stokes equations when its particles carry momentum on a sufficiently symmetric lattice and its collisions conserve mass and momentum. In that setting viscosity is computed from the collision rules rather than put in, and turbulence appears at high Reynolds number. This note takes the mechanistic route.

2.2 The equations as weighted rewrites

The path of least resistance is to stop treating the monoid equations as a quotient and to treat them as weighted rewrites: commutativity and associativity fire, at a rate. Mixing then becomes a process rather than an assumption, and the well-mixed solution becomes the limit in which the structural rewrites are infinitely fast relative to communication. Diffusion, locality and a phenomenological viscosity (a graded clause on the swap rewrite, by Stokes–Einstein) all follow at once.

The costs are real. The weighted-GSLT note holds that structural rules are addressing and incur no primitive cost [20]; the unitary-equations note places the coherent part of the dynamics in the equations [19], so demoting equations to reductions moves them to the measuring side and would decohere mixing under $\mathbb{V} = \mathbb{C}$. The commutativity that forces the permutation symmetry behind the permanent in the quantum note [7] is lost once positions are distinguishable. And the geometry the oriented equations induce is the wrong one: orienting commutativity alone gives a one-dimensional exclusion process, in which there is no turbulence; orienting associativity as well gives tree rotations, whose neighbourhoods are closer to hyperbolic than Euclidean.

2.3 Interaction means close

The constraint the author of this note placed on the geometry is that it must *emerge from the rewrite rules*: two molecules are close to the degree that they interact. On this reading the well-mixed solution is not a geometry but its absence — the complete interaction graph, of no finite dimension. Geometry can appear only when the rewrites make interaction sparse and repeated. Distance is then a statistic of the dynamics, such as commute-time distance on the rate-weighted interaction graph, and dimension is read off spectrally or from ball growth, as in the reconstruction of chromosome conformation from contact frequencies [9] or the condensation of space from a complete graph in quantum graphity [8]. Viscosity acquires an intrinsic definition as the lifetime of neighbourhoods, in the spirit of Maxwell's relation between a liquid's viscosity and its relaxation time. This completes, on the spatial side, the construction of [18], where the causal order of a spacetime is generated by the dynamics but nearness is supplied by a contact constructor. The hazard is circularity: if closeness is *only* interaction then "only near things interact" is a tautology, and something must make interaction self-reinforcing.

2.4 Bonds

With the restriction operator, $(\texttt{new}\ x)(P(x) \mid Q(x))$ is a bond, but an unbreakable one. Scope extrusion, $(\texttt{new}\ x)P \mid Q \equiv (\texttt{new}\ x)(P \mid Q)$ when $x \notin \mathrm{fv}(Q)$, read as a rewrite, is bond formation; its reverse carries the same side condition, so once Q holds x it can never leave,

and nothing outside can ever guess x. The bond has infinite binding energy. One natural way to break it is a collision that replaces the bound name by its image under the encoding of restriction into rho: in the repaired encoding [16] that image is a quoted address allocated by Fresh_2, an ordinary name. Breaking the bond then changes nothing inside $P \mid Q$ and everything about who else may reach x. Re-bonding is not the inverse, since rho cannot re-privatise a public name.

This line of thought led directly to the question of what a collision *is*, and the answer it found is the subject of the rest of the note. The bond returns in Section 8.1, where collision itself forms one.

3 Locations, races and critical pairs

Definition 3.1 (Location)**.** A *location* in a term T is a splitting $\langle K, t \rangle$ of T into a one-hole context K and a term t with $K[t] \equiv T$. The location is pointed to by $K[t]$.

A reduction rule acts at a location. Writing a receipt as a context, two messages on its channel give the familiar shape

$$P \mid K \mid Q \rightsquigarrow K[P] \mid Q \qquad \text{or} \qquad P \mid K \mid Q \rightsquigarrow P \mid K[Q],$$

which does not type-check — K is a context and P, Q are terms — but works as a schema for instances that do. The instance of interest is

$$x!(P) \mid \texttt{for}(y \leftarrow x)R \mid x!(Q) \rightsquigarrow R\{@P/y\} \mid x!(Q) \quad \text{or} \quad x!(P) \mid R\{@Q/y\}.$$

Remark 3.2 (A race is a critical pair). The two redexes overlap on the receipt, so the two reducts form a critical pair in the sense of Knuth and Bendix [12]. The pair is not joinable, and it is not meant to be: the race is the semantics. Completion would join the peak or orient it into a new rule. The construction of this note does neither. It *names the peak*: a new kind of interaction is introduced for the case in which the race is not resolved because both bidders arrive.

There are two shapes of overlap. In the one displayed, two data contend for one receipt. In the dual, $\texttt{for}(y \leftarrow x)R \mid x!(P) \mid \texttt{for}(z \leftarrow x)S$, two receipts contend for one datum. These are the two contention cases of the contention-set normalisation in the quantum note [7]: candidates conflict when they share a datum or a linear receipt. This note develops the first shape, which is the one a physical collision has, and returns to the second in Section 9.

4 The collision kernel

4.1 Collision

A collision is two terms racing to the same location in which, instead of one winning, both get there, and their arrival causes an interaction. The first form of the idea introduces an interaction term $P \circ Q$ and an augmented reduction relation $\rightsquigarrow'$:

$$x!(P) \mid \texttt{for}(y \leftarrow x)R \mid x!(Q) \rightsquigarrow' R[P \circ Q].$$

As written, the right-hand side absorbs both messages and leaves nothing on x. That is not scattering but *association*: P and Q end up held together inside R by the collision. It is the bond of Section 2, formed this time by a three-body event, and it is recovered below as a special case.

4.2 Scattering

To obtain scattering, refine the interaction so that $P \circ Q$ is multivalued, $[S_1, \ldots, S_n]$, and give R several holes — these are the locations at which the S_i land. The assignment of products to holes is then the remaining question, and it is sidestepped by letting the interaction return the assignment itself.

Definition 4.1 (Scattering kernel)**.** Let R have free name variables $y_1, \ldots, y_n$. A *scattering kernel* for R is a map κ from pairs of processes to finite sets of substitutions of the form $\sigma = \{@S_1/y_1, \ldots, @S_n/y_n\}$. The *collision rule* is

$$x!(P) \mid \texttt{for}(y_1 \ldots y_n \leftarrow x)R \mid x!(Q) \;\rightsquigarrow'\; R\sigma \qquad (\sigma \in P \circ Q := \kappa(P, Q)),$$

and each σ is weighted by the receipt's graded where-clause.

The collision rule has the shape of COMM. Ordinary communication yields $R\{@P/y\}$, a substitution fixed by the match; collision yields $R\sigma$, a substitution computed from both arrivals. So $\rightsquigarrow'$ *generalises* communication rather than adding a new term former beside it: the scattering is σ, and the locations are the variables. It also drops into the graded clause unchanged. The graded clause weights candidate matches, and a candidate match *is* a substitution; if κ returns a family of substitutions, the clause weights them exactly as it weights any family of candidates.

Remark 4.2 (Geometry and dynamics separate). The receiver fixes *where* products can land, through its arity and its holes; the kernel fixes *what* lands. This is how a scattering experiment is organised — a beam, a target, and detectors at fixed angles — and it is the separation that lets a later section put momentum on the holes.

4.3 Four kernels

Example 4.3 (Association)**.** One hole, $\kappa(P, Q) = \{\{@(P \mid Q)/y\}\}$. Both arrivals end up together in one hole.

Example 4.4 (The race, recovered)**.** Two holes, with the receipt's continuation $R' = R \mid x!(*z)$, and $\kappa(P, Q) = \{\{@P/y, @Q/z\}, \{@Q/y, @P/z\}\}$. Then $R'\{@P/y, @Q/z\} = R\{@P/y\} \mid x!(*@Q) \equiv R\{@P/y\} \mid x!(Q)$, which is the COMM reduct exactly; the "loser" is re-emitted exactly because $*@Q \equiv Q$. The identity kernel therefore reaches the same outcomes as the race. It does not reach them at the same *rate*: its weight adds to COMM's, a point taken up in Section 6.

Example 4.5 (Exchange)**.** Two holes; if $P \equiv A \mid B$ and $Q \equiv C \mid D$, a kernel containing $\{@(A \mid D)/y_1, @(C \mid B)/y_2\}$ exchanges parts. Elastic scattering with exchange is crossover on quoted code.

Example 4.6 (Fission)**.** $n > 2$ holes; κ decomposes $P \mid Q$ into n fragments. The holes may be binder positions that R uses or emission positions $x_i!(*y_i)$, in which case the fragments scatter back into the medium in n directions.

4.4 Momentum without coordinates

Lattice-gas automata carry momentum as *which outgoing direction* a particle travels, and their collisions conserve mass and momentum by redistributing particles among directions [2]. The holes of a receiver are those directions. Label each hole y_i with a direction e_i and give each process a mass $m(-)$. Conservation is then a property of the kernel:

$$m(P) + m(Q) = \textstyle\sum_i m(S_i), \qquad p_{\text{in}} = \textstyle\sum_i e_i\, m(S_i).$$

Hydrodynamics needs no stated geometry beyond this — only a kernel respecting these constraints over a sufficiently symmetric set of holes. In graded terms the two physical ingredients become two clause disciplines: conservation laws are crisp guards, which zero the violating substitutions and cost nothing [1]; the cross-section is the graded value on the substitutions that remain.

5 Destructuring patterns and the canonical kernel

5.1 Patterns

Assume term-level destructuring on the left of the arrow: in $\texttt{for}(@\pi \leftarrow x)R$ the pattern π ranges over the term constructors of processes, some of whose arguments may be variables, together with process variables, constants and the parallel composition of patterns. Matching is modulo the monoid laws of $|$, so $\mathrm{match}(\pi, T)$ is a finite *set* of substitutions. A calculus with such receipts has continuations with multiple holes by default, which removes the arity problem of Definition 4.1.

5.2 The canonical kernel

Definition 5.1 (Canonical kernel)**.** For a receipt with pattern π, the canonical k-body kernel is matching against the composite of the arrivals:

$$P_1 \circ \cdots \circ P_k \;:=\; \mathrm{match}(\pi,\; P_1 \mid \cdots \mid P_k).$$

The kernel calculus rho° has, for each $k \geq 1$, the rule

$$x!(P_1) \mid \cdots \mid x!(P_k) \mid \texttt{for}(@\pi \leftarrow x)R \;\rightsquigarrow_k\; R\sigma \qquad (\sigma \in P_1 \circ \cdots \circ P_k),$$

weighted $w_k \cdot g(\sigma)$, where g is the receipt's graded where-clause and $w_k \in \mathbb{R}_{\geq 0}$ is the k-body cross-section. Persistent receipts remain.

No new primitive is required: AC matching is already multivalued, so the kernel returns a family of substitutions for free, and the receiver's pattern is the scattering geometry — its variables are where the products land. $\rightsquigarrow_1$ is COMM. The relation $\rightsquigarrow'$ of Section 4 is $\bigcup_{k\geq 2} \rightsquigarrow_k$.

Proposition 5.2 (The virial series)**.** *Let a site hold one receipt with pattern π and messages $x!(M_1), \ldots, x!(M_n)$. The total propensity of the site is*

$$a \;=\; \sum_{k\geq 1} w_k \sum_{\substack{S\subseteq\{1..n\}\\ |S|=k}} \;\sum_{\sigma\in \mathrm{match}(\pi, \prod_{i\in S} M_i)} g(\sigma).$$

For a bare-variable pattern $\pi = y$ and $g \equiv 1$ this is $a = \sum_k \binom{n}{k} w_k$.

Proof. Candidates are counted as selections from the multiset of messages, so a k-body candidate is a k-subset of messages together with a substitution; the bare variable matches any composite in exactly one way. □

The coefficient w_k plays the role of the k-th virial coefficient: it is the correction that k-body encounters make to the ideal gas, and its contribution grows as the k-th power of the occupancy. The rho calculus is the truncation at $k = 1$. Table 1 gives the $k \leq 2$ rates computed by the generic enumerator (experiment E2).

n	$a, w_2 = 0$	$a, w_2 = 0.1$	$n + \binom{n}{2} \cdot 0.1$
1	1.000	1.000	1.000
2	2.000	2.100	2.100
5	5.000	6.000	6.000
10	10.000	14.500	14.500
20	20.000	39.000	39.000

Table 1: Site propensity for a bare-variable receipt with $w_1 = 1$, computed by enumerating k-subsets and matches.

5.3 The pattern decides whether a receipt can collide

Proposition 5.3 (Shape)**.** *Let every message payload be non-zero.*

(i) If π is headed by a constructor other than $|$, then $\mathrm{match}(\pi, P_1 \mid \cdots \mid P_k) = \emptyset$ *for every $k \geq 2$.*

(ii) If $\pi = \pi_1 \mid \cdots \mid \pi_m$ with every π_j headed by a constructor other than $|$ (so no π_j is a variable or matches 0), then $\mathrm{match}(\pi, P_1 \mid \cdots \mid P_k) = \emptyset$ *for $k > m$.*

(iii) If π contains a process variable at top level, the collision order is unbounded.

Proof. (i) A composite of $k \geq 2$ non-zero processes is headed by $|$ up to congruence. (ii) A composite of k non-zero processes has at least k top-level components, and a match distributes them among the π_j; each π_j, being headed by a constructor other than $|$, absorbs exactly one. So the composite has exactly m components and $k \leq m$. (iii) A variable absorbs any composite. □

So receipts stay unmarked in the syntax, but the pattern marks them. A pattern headed by a constructor, such as $a!(u)$, is ideal-gas by construction. Only a top-level $|$ or a bare variable can absorb more than one arrival, and only a variable makes the order unbounded.

Remark 5.4 (Fusion is what the pattern says)*.* A bare-variable receipt $\texttt{for}(y \leftarrow x)R$ says "whatever is at this location". With $w_2 > 0$ it therefore fuses requests: two arrivals become $\{@(P \mid Q)/y\}$. This is not a default imposed by the kernel; it is what the pattern literally says. A program that does not want fusion writes a constructor-headed pattern.

5.4 Conservation as a syntactic criterion

Let the mass $m(P)$ be the number of non-zero, non-parallel components of P (any additive cost measure will do).

Proposition 5.5 (Mass at the pattern)**.** *Let $\pi = c_\pi \mid v_1 \mid \cdots \mid v_j$, with c_π a parallel composition of closed constants and the v_i distinct variables (so π is linear and contains no wildcard). Then for every $\sigma \in \mathrm{match}(\pi, P_1 \mid \cdots \mid P_k)$,*

$$\textstyle\sum_i m(P_i) \;=\; m(c_\pi) + \sum_v m(*\sigma(v)),$$

A wildcard in place of a variable discards the mass it matches. The same accounting applies component by component to patterns whose components are constructors with variable arguments, such as $a!(u)$.

All incoming content therefore lands either in a hole or in the pattern's constants. Constants are absorbed by the receiver, which acts as a catalyst unless R re-emits them; for constants-only patterns, such as those of Section 8, conservation is the syntactic check that R emits the mass its pattern absorbs.

Remark 5.6 (Bound and collided look the same). One message $x!(A \mid B)$ and two messages $x!(A)$, $x!(B)$ produce the same substitution. A receiver cannot tell a pre-bonded molecule from a collision complex; only the rates differ, first order against second. Physically this may be right — the location holds the same stuff — but it means a bond is visible only kinetically, never structurally. A collision constructor in the pattern language would distinguish the two at the cost of a new pattern former.

6 Unmarked receipts

Should the kernel mark collision-capable receivers in its syntax? Marking keeps the translation local and makes the identity on rho trivial; leaving receipts unmarked is closer to physics, in which every molecule can collide. We leave them unmarked and count the cost.

Proposition 6.1 (The ideal gas)**.** *If $w_k = 0$ for all $k \geq 2$, the outcome-weighted semantics of rho$^\circ$ is that of graded rho, outcome by outcome.*

Proof. Only $\rightsquigarrow_1$ has non-zero weight, and $\rightsquigarrow_1$ is COMM. □

Experiment E2 confirms the statement on 2,000 random instances with discrepancy exactly 0. Marking therefore returns, but semantically: a receipt with zero cross-section is a receipt of the ideal gas, and the translation of Section 7 will be the identity on it. The costs of the unmarked choice are the following.

Kinetics change, and sometimes support. For the identity kernel of Example 4.4 collisions reach only COMM outcomes, so crisp support survives; for the canonical kernel a pattern that can absorb several arrivals reaches new outcomes as well (Remark 5.4). In either case, under $\mathbb{V} = \mathbb{R}_{\geq 0}$ collision weight adds to the propensity of a site, and it adds superlinearly in occupancy. Within one site this reweights outcomes; across sites it reweights the jump chain, boosting crowded sites (Table 2). Rates are conserved only in the limit $w_k \to 0$ for $k \geq 2$, or equivalently the dilute limit. This is physically correct: it is the departure of a real gas from an ideal one as density grows.

w_2	P(site with 2 messages fires)	P(site with 20 messages fires)
0.00	0.090909	0.909091
0.01	0.084065	0.915935
0.10	0.051095	0.948905
1.00	0.014085	0.985915

Table 2: Jump-chain probability of the next firing, two bare-variable sites, $w_1 = 1$.

The translation gets expensive. Every collidable receipt acquires a lock and a family of self-joins (Section 7); a persistent receipt needs a fresh lock at every firing, so the cost-accounted translation charges one Fresh_2 rendezvous per firing. Joins compound it: in a k-way join each component could collide independently, giving 2^k branches, which the dilute approximation — at most one collision per firing — reduces to $k + 1$.

Unbounded order. By Proposition 5.3(iii) a pattern with a top-level process variable admits collisions of every order. The series of Proposition 5.2 does not stop and the translation would be infinite. A truncation order K is required; it is the dilute-gas approximation, stated as a parameter.

The coherent instance breaks. Under $\mathbb{V} = \mathbb{C}$ the one-body and k-body branches *interfere* wherever they reach the same outcome, so every communication acquires a phase-sensitive correction and can be cancelled outright. This note is scoped to $\mathbb{R}_{\geq 0}$.

What unmarked buys. There is no marking to preserve under reduction. The choice-formulae note shows that a static annotation can be mailed in through reflection: a received process dropped by $*y$ brings its own receipts into the receiver's code [1]. With every receipt treated alike, that is not an issue.

7 The map back to rho

The presentation is a kernel that makes the intuitions clear, followed by a map from the kernel back to rho without it. The ordinary branch and the collision branches must compete for the same messages, and rho has no summation, so each receipt becomes a family of atomic joins sharing a lock.

7.1 Splitting a pattern across arrivals

In a k-fold self-join each join component sees one message, so the pattern must be split across the arrivals. A variable may *straddle*: in $\mathrm{match}(\pi, P \mid Q)$ a variable u may bind $A \mid B$ with A drawn from P and B from Q.

Definition 7.1 (Split)**.** A k-split of $\pi = c_\pi \mid v_1 \mid \cdots$ assigns each occurrence of a constant in c_π to one of k components and replaces each variable v by fresh variables $v^1, \ldots, v^k$, one per component. Component i is the pattern $\pi_i^s = c_i^s \mid v_1^i \mid \cdots$. $\mathrm{Split}_k(\pi)$ is the set of k-splits; for a pattern whose constants have multiplicities c_a it has $\prod_a \binom{c_a+k-1}{k-1}$ elements.

Lemma 7.2 (Recombination)**.** *Let $\theta_v := @(*v^1 \mid \cdots \mid *v^k)$. If v^i is bound to $@S^i$ then θ_v becomes $@(S^1 \mid \cdots \mid S^k)$.*

Proof. Substitution gives $@(*@S^1 \mid \cdots \mid *@S^k)$, and $*@S \equiv S$ inside the quote, since names are equal when their quoted processes are congruent. □

The equation $*@P \equiv P$ therefore carries the correctness of the translation.

7.2 The translation

Definition 7.3 (Translation)**.** For a linear receipt and truncation order K,

$$[\![\texttt{for}(@\pi \leftarrow x)R]\!]_K \;=\; \texttt{new}\; l\; \texttt{in}\; \Big(l!(0) \;\Big|\; \prod_{k=1}^{K} \prod_{s \in \mathrm{Split}_k(\pi)} \texttt{for}(@\pi_1^s \leftarrow x \,\&\, \cdots \,\&\, @\pi_k^s \leftarrow x \,\&\, _ \leftarrow l\; \texttt{where}\; \omega_k)\; R\{\theta_v/v\}_v \Big),$$

with clause value

$$\omega_k \;=\; \frac{w_k\, g(\sigma)}{\mu(P_1, \ldots, P_k; \sigma)},$$

where σ is the recombined substitution and μ is the fibre multiplicity of Definition 7.4. The lock is allocated by Fresh_2 when the target is rho without restriction [16]. A persistent receipt reinstalls its lock, freshly allocated, at every firing.

Atomic joins remove the partial-commitment problem that defeats encodings of guarded choice through a lock, since a join commits all at once. Gillespie's selection counting then gives the k-body branches their combinatorics automatically: the second-order kinetics of collision are a consequence of the translation, not a stipulation. The losing branches are left blocked on a consumed private lock; they are behaviourally inert, but correctness is stated up to this garbage.

7.3 The fibre multiplicity

The naive expectation is that each translated candidate should carry the kernel's weight $w_k g(\sigma)$, perhaps divided by the symmetry factor $k!$ for the orderings of the join. Both are wrong. A single kernel outcome is reached by many translated candidates: every ordering of the k messages, every split, and every decomposition of each straddling variable's binding across the components. When atoms are indistinguishable, several splits reach the same outcome.

Definition 7.4 (Fibre multiplicity)**.** For messages $P_1, \dots, P_k$ and an outcome σ, let $n_{i,a}$ be the number of occurrences of atom a in P_i, c_a that in c_π, and $s_{v,a}$ that in $*\sigma(v)$. Let T_a be the number of non-negative integer matrices with rows indexed by i, columns indexed by the roles $\{c\} \cup \{v\}$, row sums $n_{i,a}$ and column sums $(c_a, s_{v,a}, \dots)$. Then

$$\mu(P_1, \dots, P_k; \sigma) \;=\; k! \prod_a T_a.$$

Theorem 7.5 (Correspondence)**.** *Fix a site and a truncation order K. For every outcome — a set of consumed messages together with a substitution — the total weight assigned by the translation equals the weight $w_k g(\sigma)$ assigned by* rho$^\circ$.

Proof sketch. Translated candidates reaching a fixed outcome are in bijection with pairs of an ordering of the consumed messages and, for each atom type, a matrix distributing the occurrences of that atom in each message among the roles; a matrix is admissible exactly when its column sums reproduce the outcome. There are therefore μ of them, each weighted $w_k g(\sigma)/\mu$. Conversely every kernel outcome is reached from every ordering, because $\pi\sigma \equiv P_1 \mid \cdots \mid P_k$ makes each P_i a sub-multiset of $\pi\sigma$. □

Remark 7.6 (μ is admissible). μ depends only on the messages a candidate consumes and on its own bindings, both of which a where-clause sees. The correction is therefore expressible in the target, and the translation stays within graded rho. It is the outcome-fibre normalisation of the quantum note [7] met again in a different setting: weights must be summed over the fibre of an outcome, not over the derivations that reach it.

Empirical finding 7.7 (E1)**.** Over 3,000 random sites (1,820 with a non-empty kernel), with up to four messages over three atom types, patterns with up to two constants and two variables, and $K = 3$:

- with weight $w_k g/\mu$ the translation agrees with the kernel outcome by outcome, worst relative discrepancy 1.6×10^{-15}, with identical supports; the largest multiplicity met is $\mu = 108$;
- with weight $w_k g/k!$ the total weight is overcounted by a factor of 1.281 on average and up to 5.162;
- with weight $w_k g$ the overcount is 3.639 on average and up to 24.32.

Proposition 7.8 (Identity at zero cross-section)**.** *If $w_k = 0$ for $k \geq 2$ the lock and all collision branches may be omitted, and $[\![\texttt{for}(@\pi \leftarrow x)R]\!]_K = \texttt{for}(@\pi \leftarrow x)R$. In particular the translation is the identity on every* rho *program whose receipts have zero cross-section.*

Proof. With a single branch the lock never decides anything, and $\mu = 1$ for $k = 1$. □

8 Worked examples

8.1 Small kernels

Association is bonding. The receipt

```
for( @y <- x ){ bound!( *y ) }
```

with $w_2 > 0$ turns two arrivals $x!(P)$, $x!(Q)$ into $\texttt{bound}!(P \mid Q)$: P and Q now travel together. Unlike $(\texttt{new}\ z)(P \mid Q)$ the bond is breakable, by a dissociation receipt whose pattern recognises the parts:

```
for( @(a!(u) | b!(v)) <= bound ){ x!( a!(*u) ) | x!( b!(*v) ) }
```

By Remark 5.6 the receiver on `bound` cannot tell a bound pair from a pair that happens to arrive together; it sees the same substitution, at a different rate.

Exchange. Molecules are pairs $\texttt{left}!(A) \mid \texttt{right}!(B)$. The receipt

```
for( @( left!(u1) | right!(u2) | left!(u3) | right!(u4) ) <- x ){
  x!( left!(*u1) | right!(*u4) ) | x!( left!(*u3) | right!(*u2) )
}
```

cannot fire on one message, which has two parts against a four-part pattern (Proposition 5.3(ii)); it is a collision-only receipt. On two messages AC matching returns four substitutions, according to which `left` is bound to `u1` and which `right` to `u2`. Two of them return the input pair unchanged (elastic scattering without exchange) and two exchange partners; the graded clause decides the ratio. Mass is conserved: the pattern is linear and wildcard-free, and the four sends it absorbs are re-emitted as four sends.

Fission.

```
for( @(u1 | u2 | u3) <- x ){ x1!( *u1 ) | x2!( *u2 ) | x3!( *u3 ) }
```

scatters the content of the arrivals in three directions. Its pattern has top-level process variables, so by Proposition 5.3(iii) its collision order is unbounded and a truncation order must be chosen.

8.2 Collisional activation: Lindemann–Hinshelwood

A molecule A cannot decompose until a collision with a bath molecule M has activated it [13, 10]. On one location x the laws are four persistent receipts:

```
for( @(A | M)  <= x ){ x!(Astar) | x!(M) }   // activation,   k = 2,
   weight k1
for( @(Astar | M) <= x ){ x!(A) | x!(M) }    // deactivation, k = 2,
   weight km1
for( @Astar <= x ){ x!(B) }                  // decomposition, k = 1,
   weight k2
for( @B <= x ){ x!(A) }                      // recycling,     k = 1,
   weight r
```

The first two patterns are constants-only with two components, so by Proposition 5.3(ii) they fire only as two-body collisions, and the generic enumerator gives them propensity $k_1 n_A n_M$ and $k_{-1} n_{A^*} n_M$ (experiment E3: 400 random site configurations, discrepancy 0). In the ideal

gas, $w_2 = 0$, activation is impossible: the pattern $A \mid M$ matches no single message. In plain rholang one would have written the join $\texttt{for}(@A \leftarrow x \;\&\; @M \leftarrow x)$, and that is precisely what the translation of Section 7 produces.

Each molecule of A moves independently among A, A^* and B, so the effective unimolecular rate constant $k_{\text{uni}} = J/\langle n_A \rangle$ satisfies the Lindemann law exactly:

$$k_{\text{uni}} \;=\; \frac{k_1 k_2\, n_M}{k_{-1} n_M + k_2},$$

second order ($k_1 n_M$) at low bath density, where collisions limit the rate, and first order ($k_1 k_2 / k_{-1}$) at high density, where decomposition does. Table 3 gives the fall-off curve measured by stochastic simulation of the kernel.

n_M	k_{uni} measured	exact	low-density limit	high-density limit
1	0.907 ± 0.005	0.909	1.0	10.0
3	2.309 ± 0.016	2.308	3.0	10.0
10	4.981 ± 0.019	5.000	10.0	10.0
30	7.538 ± 0.061	7.500	30.0	10.0
100	9.053 ± 0.163	9.091	100.0	10.0
300	9.810 ± 0.150	9.677	300.0	10.0
1000	9.862 ± 0.116	9.901	1000.0	10.0

Table 3: Fall-off of the unimolecular rate with bath density. $k_1 = k_{-1} = 1$, $k_2 = 10$, $r = 50, 50$ molecules of A; each measurement is the mean $\pm$ standard deviation of three runs of 10,000 decompositions.

8.3 A lattice gas whose viscosity comes from the kernel

This is the example that is more than a toy. A shear wave is a sinusoidal profile of x-momentum varying in y; in a fluid it decays as $e^{-\nu k^2 t}$, and the decay rate measures the kinematic viscosity ν. We write a lattice gas in rho°, measure ν, and find that it is produced by the collision kernel — and that the kernel's two-body term is not enough.

The program. Sites are channels s on an $L \times L$ torus; a particle moving in direction d is a message $s!(p_d)$. At every site the laws are

```
for( @p_d <= s ){ (s + e_d)!(p_d) }                                  // stream, k
    = 1, h
for( @(p_d | p_{d+3}) <= s ){ s!(p_{d+1}) | s!(p_{d+4}) }    // pair,  k
   = 2, w2/2
for( @(p_d | p_{d+3}) <= s ){ s!(p_{d-1}) | s!(p_{d+2}) }    // pair,  k
   = 2, w2/2
for( @(p_0 | p_2 | p_4) <= s ){ s!(p_1) | s!(p_3) | s!(p_5) } // triple,
   k = 3, w3
for( @(p_1 | p_3 | p_5) <= s ){ s!(p_0) | s!(p_2) | s!(p_4) } // triple,
   k = 3, w3
```

with six directions e_d at angles $60°d$ on a hexagonal lattice (FHP, [2]), indices mod 6. The square-lattice variant (HPP, [11]) has four directions and the pair law $(E, W) \leftrightarrow (N, S)$. Unlike the classical automata the model is asynchronous, in continuous time, and without exclusion: a site may hold any number of particles in each direction. Geometry here is supplied by the namespace — the site names — not grown from interaction; Section 9 returns to this.

All collision patterns are constants-only, so the collision orders are fixed ($k = 2$ and $k = 3$) by Proposition 5.3(ii), the propensities are $\frac{w_2}{2} n_d n_{d+3}$ per rotation and $w_3 n_0 n_2 n_4$, $w_3 n_1 n_3 n_5$ (experiment E3, discrepancy 0), and every law conserves mass and momentum: a head-on pair and its rotation both have zero momentum, as do the two triples of directions 120° apart.

Proposition 8.1 (Line invariants). *(i) In the HPP kernel the quantity $\sum_{s \in r}(n_E(s) - n_W(s))$ is invariant for every row r.*

(ii) In the FHP kernel with $w_3 = 0$, for every horizontal line r the quantity $\sum_{s \in r}(n_0(s) - n_3(s))$ is invariant.

(iii) The triple laws change $n_0 - n_3$ by ∓ 2 at a site, so for $w_3 > 0$ statement (ii) fails.

Proof. A pair collision removes one particle from each of two opposite directions and adds one to each of two others, so it preserves every difference $n_d - n_{d+3}$. Particles in directions 0 and 3 (HPP: E and W) stream along their line and never leave it; no other direction contributes to the difference. For (iii), $(0, 2, 4) \to (1, 3, 5)$ removes a 0 and adds a 3. □

In the initial condition used here the occupation of direction d at line r is Poisson with mean $f(1 + a\cos(60°\,d)\sin(2\pi r/L))$ (for HPP, $f(1 + 2a\cos(90°\,d)\sin(2\pi r/L))$). The observable is the projected x-momentum $A(t) = \sum_{s,d} n_d(s)\cos(60°\,d)\sin(2\pi r(s)/L)$, whose expected initial value is proportional to $fa\sum_d \cos^2(60°\,d) = 3fa$. Directions 0 and 3 contribute $2fa$ of it, so Proposition 8.1(ii) predicts that without triple collisions the shear wave can decay to no less than $\frac{2}{3}$ of its initial amplitude. For HPP the whole of A is invariant.

Empirical finding 8.2 (E5, invariants). With $L = 16$, $f = 2$, $a = \frac{1}{2}$ and six runs averaged, the mean of $A(t)/A(0)$ over the second half of a run of length 60 is

kernel	late-time $A/A(0)$
HPP, pairs ($w_2 = 1$)	1.0000
FHP, no collisions (ideal gas)	0.6470
FHP, pairs only ($w_2 = 1$, $w_3 = 0$)	0.6556
FHP, pairs and triples ($w_2 = 1$, $w_3 = \frac{1}{2}$)	0.1674

against the predicted 1 for HPP and $\frac{2}{3}$ for FHP without triples.

The HPP value is exact because the whole observable is a sum of line invariants. The FHP ideal gas sits at the same two thirds as the pair kernel for a different reason: without collisions the particles in the four oblique directions keep their directions, and the randomness of asynchronous streaming spreads and dephases their contribution, while directions 0 and 3 never leave their lines. Two-body collisions, which are what a reader would write first, therefore buy nothing for this mode: they relax every oblique contribution that free streaming would have dephased anyway, and leave the invariant untouched. The three-body term — the third virial coefficient of Proposition 5.2 — is what makes the shear wave decay. In the language of the lattice-gas literature this is the familiar observation that pair collisions alone leave spurious conservation laws, and that triple collisions are added to FHP to remove them [3]; here it appears as a statement about which terms of the virial series a kernel keeps.

Viscosity. With pair and triple collisions the shear wave decays. Let $\lambda = w_2 f$ be the collision parameter: a particle meets a head-on partner at rate $\propto w_2 f$. Triple weights are tied to it as $w_3 = w_2/f$, so that the per-particle triple rate also scales as λ. Kinetic theory predicts that the viscosity is proportional to the mean free path, hence to $1/\lambda$, plus a contribution from the lattice itself.

f	w_2	$\lambda = w_2 f$	Γ	$\nu = \Gamma/k^2$	events
2	0.25	0.5	0.0573	0.2784	1,533,786
2	0.5	1	0.0414	0.2013	1,973,516
1	1	1	0.0428	0.2082	982,087
2	1	2	0.0382	0.1857	2,857,231
2	2	4	0.0351	0.1706	4,618,405
4	1	4	0.0272	0.1320	9,332,797
2	4	8	0.0304	0.1480	8,124,015

Table 4: Decay rate and kinematic viscosity of the shear wave, FHP kernel with pair and triple collisions ($w_3 = w_2/f$), $L = 16$, $k^2 = 0.2056$, six runs averaged per row.

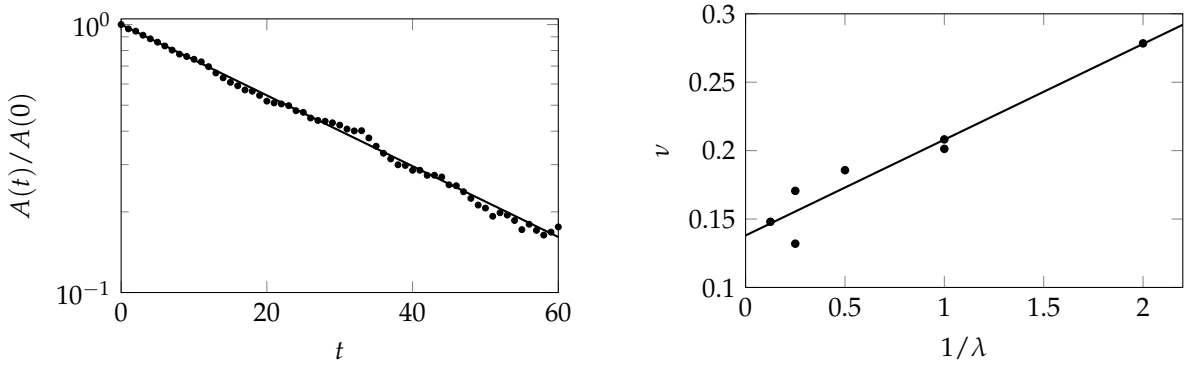


Figure 1: Left: the averaged shear wave at $\lambda = 8$ with its fitted exponential. Right: ν against $1/\lambda$, with the least-squares line $\nu = \alpha/\lambda + \beta$.

Empirical finding 8.3 (E5, E6, viscosity)**.** Table 4 and Figure 1. The viscosity falls from 0.278 at $\lambda = \frac{1}{2}$ to 0.148 at $\lambda = 8$. A least-squares fit gives

$$\nu \;=\; \frac{\alpha}{\lambda} + \beta, \qquad \alpha = 0.070, \quad \beta = 0.138,$$

with largest residual 0.024. Replicates with fresh seeds (E6) put the standard error of a single row at 0.023–0.027 for $\lambda = 1$, 0.010 for $(f, w_2) = (2, 2)$ and 0.005 for $(4, 1)$.

The $1/\lambda$ term is the kinetic-theory term: the viscosity of a gas is proportional to its mean free path, and the mean free path is inversely proportional to the collision rate, which the kernel's cross-section sets. The constant β remains as $\lambda \to \infty$ and is a contribution of the lattice and of asynchronous streaming; it is not derived here, and it is large — at the collision parameters reachable in this study it dominates. Whether ν depends on w_2 and f only through λ is a sharper test. At $\lambda = 1$ the two ways of reaching it agree within error (0.201 and 0.208; replicates 0.225 and 0.194). At $\lambda = 4$ they do not: the denser gas is less viscous (0.132 against 0.171; replicates 0.153 and 0.179), a difference of about two standard errors in each pair. The density therefore enters through the lattice term as well, and Maxwell's density-independence of the dynamic viscosity (mass density times ν) is not reproduced by this model at this precision. The viscosity *is* a property of the collision kernel — without the three-body term the mode does not relax at all, and with it the viscosity moves with the cross-section — but the model is a lattice gas, and the lattice has a say.

9 What is shown, what is not, and what is open

Shown. The kernel calculus rho$^\circ$ with the canonical kernel; COMM as its one-body rule and rho as its ideal gas (Proposition 6.1); the virial series of site propensities (Proposition 5.2); the shape criterion (Proposition 5.3) and mass at the pattern (Proposition 5.5); the translation back to rho and its correspondence with the kernel once the fibre multiplicity is included (Theorem 7.5, with a proof sketch and Finding 7.7); identity at zero cross-section (Proposition 7.8); the line invariants of the two-body lattice kernels (Proposition 8.1).

Measured, not proved. The Lindemann fall-off; the decay of the shear wave and the dependence of the viscosity on the collision parameter. These are statements about finite simulations, with the finite-size and statistical caveats that implies.

Obligation 9.1 (A full proof of correspondence)**.** Theorem 7.5 is stated at a site and proved by a counting argument. The operational correspondence for whole terms — completeness and soundness up to administrative steps on locks and up to garbage, together with weight preservation on the jump chain — remains to be written out, including the persistent case in which each firing allocates a fresh lock by Fresh_2.

Obligation 9.2 (Emergent geometry)**.** The lattice gas takes its geometry from the site names. The aim stated in Section 2 is a geometry grown from interaction, with distance a statistic of the dynamics. The construction of this note supplies the ingredients — collisions are three-party events, and bonds formed by association are persistent edges of the interaction graph — but not the self-reinforcement that would make the interaction graph settle into a low-dimensional geometry rather than randomise into an expander. Measuring its spectral dimension as a function of the structural rate is the first experiment to run.

Obligation 9.3 (Turbulence)**.** Turbulence needs a high Reynolds number UL/ν: small viscosity, which the kernel supplies by a large collision parameter, together with a forcing and a system large enough for an inertial range. The asynchronous, exclusion-free FHP of this note is a starting point; its isotropy and the absence of further spurious invariants should be checked before it is driven.

Obligation 9.4 (The dual overlap)**.** Two receipts contending for one datum is the second critical-pair shape. Its collision is a collision of contexts, $R \circ S$ filled by one term. The canonical-kernel construction should extend to it, presumably by matching a composite pattern against a single payload.

Obligation 9.5 (Detailed balance and re-bonding)**.** Association and dissociation are separate laws, not inverse ones, so detailed balance holds only if their weights are tied deliberately. The relation between collision-formed bonds and the unbreakable bonds of restriction, and whether breaking the latter by exposure to its rho image belongs in the same kernel, is open.

Obligation 9.6 (Bound and collided)**.** Remark 5.6: should the pattern language acquire a collision constructor, so that a receiver can distinguish a pre-bonded molecule from a collision complex?

Obligation 9.7 (The coherent case)**.** Under $\mathbb{V} = \mathbb{C}$ the one-body and many-body branches interfere. Whether that interference is a defect or the beginning of a scattering theory with amplitudes, connecting to the permanent of the quantum note, is open.

10 Conclusion

A race in the rho calculus is a critical pair whose peak is never joined. Naming the peak gives collision: both bidders arrive, and what happens next is a substitution into the holes of the receiver. With destructuring patterns the substitution is canonical — the receiver's pattern matched against the composite of the arrivals — and the many-body terms of that construction are the virial corrections to an ideal gas that the rho calculus, read as a chemistry, has always been. The receiver's pattern decides whether it can collide, its cross-section whether it does, and the map back to rho is a lock, a family of self-joins and a weight that counts outcomes rather than derivations. In a lattice gas built this way, viscosity is produced by the kernel, and the kernel's two-body term alone does not produce it everywhere.

A The computations

All numbers are produced by `collide.py` (Python 3 standard library only). `python3 collide.py e1 e2 e3 e4 e5` reproduces them; seeds are fixed.

E1 (kernel = translation).
Random sites: two to four messages of one or two atoms over {a,b,c}; patterns with up to two constants and up to two process variables; $w_1, w_2 \in [0.2, 2]$, $w_3 \in [0, 1]$; clause $g(\sigma) = \prod_v (1 + \frac{1}{4}|\sigma(v)|)$. The kernel enumerates k-subsets and AC matches; the translation enumerates orderings, splits and per-component matches, recombines, and weights by $w_k g$, $w_k g/k!$ or $w_k g/\mu$, with μ computed by an independent contingency-table recursion.

E2 (ideal gas).
Kernel with $w_{k\geq 2} = 0$ against graded rho on 2,000 random sites; Tables 1 and 2.

E3 (propensity tables).
The generic enumerator against the closed-form propensities used by the fast simulators of E4 and E5, on 400 random site configurations.

E4 (Lindemann).
Exact stochastic simulation of the four laws; k_{uni} estimated as decompositions divided by the time integral of n_A.

E5 (lattice gas).
Exact stochastic simulation with Fenwick-tree selection over cells (streaming) and sites (collisions); $L = 16$, $a = \frac{1}{2}$, $h = 1$, $T = 60$, six independent runs per point, averaged before fitting. Γ is the least-squares slope of $\log A(t)/A(0)$ over the window in which the averaged ratio exceeds 0.2, and $\nu = \Gamma/k^2$ with $k = 2\pi/(L\sqrt{3}/2)$, the wavenumber of the mode in Cartesian coordinates.

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